College of Technology and Built Environment (CTBE)

School of Information Technology Engineering

(AI Stream)

Introduction to Robotics and Intelligent Systems

(LAB - report)

Group-3

Members-Name	**ID**
1. Abigiya Daniel	UGR/5110/15
2. Biruk Ayalew	UGR/6137/15
3. Lamesgin Desalegn	UGR/0690/15
4. Ruth Ambaw	UGR/8802/15
5. Saron Tadesse	UGR/5471/15

Instructor: Ms. Emanda Hailu

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Title : Autonomous Crop Monitoring Robot for Smallholder Farms (ROS 2 and Gazebo Simulation)

Abstract

This project presents the design and simulation of an autonomous crop monitoring robot developed using ROS 2 and Gazebo. The system integrates robot modeling, simulation, sensor integration, camera-based perception, and navigation-related behaviors in a modular ROS architecture. A simulated agricultural environment with crop rows was created, and a mobile robot equipped with a camera sensor was deployed to monitor crop conditions. Image processing techniques were applied to detect visual indicators of crop stress such as discoloration and dryness, with results published continuously as ROS topics. Experimental results demonstrate that the robot simulation, perception pipeline, and vision-based navigation behaviors operate correctly in real time. The project validates the feasibility of low-cost autonomous crop monitoring systems for smallholder farming applications in a simulated environment.

1. Introduction

Agriculture is the backbone of Ethiopia's economy, with most production relying on smallholder farmers. Crop monitoring in such settings is largely manual, making early detection of crop stress difficult and inefficient. Delayed detection of water stress, disease, or nutrient deficiency often results in reduced yield and economic loss.

Autonomous robots equipped with sensors and intelligent perception can assist farmers by providing frequent and consistent crop health monitoring. This project demonstrates a simulated autonomous crop monitoring robot capable of navigating a farm environment and analyzing crop conditions using a camera sensor. The system is implemented entirely in ROS 2 and Gazebo, serving as a foundation for future real-world deployment.

2. System Overview and Architecture

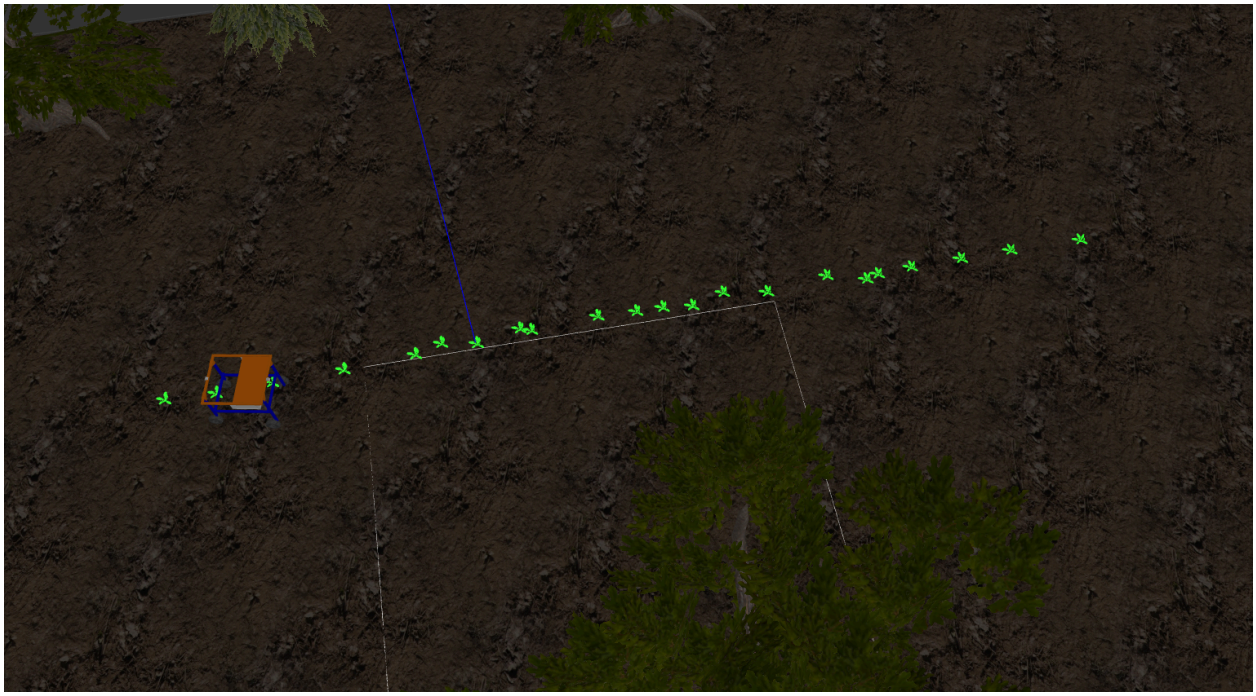
The system follows a modular ROS 2 architecture, where each functional component is implemented as an independent package. Communication between components is achieved using ROS topics and services.

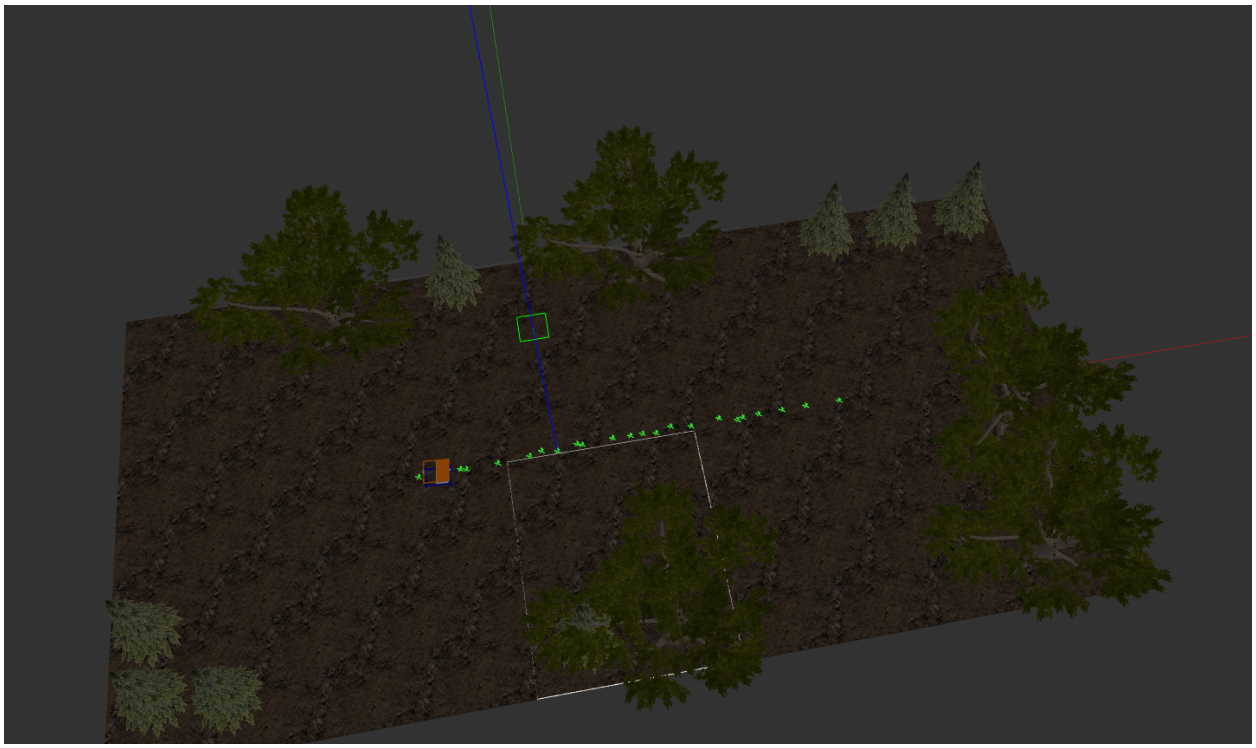
Main Packages

The repository contains the following major packages:

- **agribot_control** – Robot URDF/Xacro description, Gazebo plugins, controllers, and simulation control configuration.
- **agribot_description** – Visual and structural robot description files and meshes.
- **agribot_gazebo** – Gazebo launch files, world files, and crop/plant models.
- **agribot_croprow_generator** – Scripts and models used to generate crop-row farm environments.
- **agribot_perception** – Camera-based perception nodes for crop stress detection and vision-guided navigation.
- **agribot_joystick** – Manual control and teleoperation via joystick (optional).
- **agribot_launch** – Top-level launch files that integrate simulation, sensors, perception, and control.
- **agribot_imu, agribot_servo** – Supporting hardware-oriented packages prepared for future physical deployment.

The main demonstration is launched using `agribot_launch/demo.launch.py`, which starts the simulation, robot model, sensors, and perception nodes together.





Gazebo simulator images

3. Robot Modeling and Simulation

The robot is modeled as a differential-drive mobile platform using URDF and Xacro. The model includes wheels, chassis, sensor mounts, and a camera housing. Gazebo plugins are used to simulate wheel control and camera sensing.

A custom farm environment with crop rows was created using Gazebo world files. Crop models are placed to simulate real agricultural rows, allowing the robot to operate in a realistic monitoring scenario.

During simulation:

- The robot model loads correctly in Gazebo.
- `robot_state_publisher` publishes standard TF frames (e.g., `base_link`, sensor frames).
- The Gazebo camera plugin publishes image data continuously.

4. Perception and Crop Stress Detection

The perception subsystem is implemented in the **agribot_perception** package and consists of two main ROS nodes:

Crop Health Monitor Node

This node subscribes to the camera image topic and applies basic image processing techniques using OpenCV. The algorithm analyzes color distributions and texture features to estimate crop conditions such as greenness and dryness.

Detected crop health information is published as a ROS topic (e.g., `/agribot/crop_health`) in the form of status messages.

Vision Navigator Node

This node processes visual input to support navigation behaviors such as obstacle awareness and automatic stopping when unsafe conditions are detected. It operates concurrently with the crop monitoring node.

Experimental Results

During runtime, the system continuously published messages such as:

> **WARNING: CROP STRESS DETECTED (DRYNESS)**

These results confirm that the perception pipeline functions correctly and produces real-time diagnostic information based on camera input.

5. Navigation and Motion Control

The system supports multiple modes of movement:

- **Manual control** using joystick input via the `agribot_joystick` package.
- **Vision-guided navigation behaviors**, where the robot responds to visual cues and obstacles detected by the camera.

The robot successfully moves within the simulated environment and reacts to perception outputs. While full autonomous waypoint-based navigation using a global planner is not the primary focus of this implementation, the system demonstrates closed-loop sensing and motion behavior suitable for crop-row traversal.

6. Experimental Evaluation

All major components of the system were tested in simulation:

- Gazebo simulation launches successfully.
- Robot model spawns correctly and remains stable.
- Camera sensor publishes continuous image streams.

- Crop stress detection publishes consistent and interpretable results.
- Vision-based navigation node runs simultaneously with perception.
- ROS topics and nodes can be inspected using standard ROS 2 tools.

The observed system behavior matches the intended objectives of the project proposal.

7. How to Run the System (for Demonstration)

From the workspace root:

```
colcon build --symlink-install

source install/setup.bash

ros2 launch agribot_launch demo.launch.py gui:=true
```

To run perception nodes manually:

```
source install/setup.bash

ros2 run agribot_perception vision_navigator_node

ros2 run agribot_perception crop_monitor_node
```

To monitor crop health output:

```
ros2 topic echo /agribot/crop_health
```

8. Individual Contributions

- **Robot Modeling and Simulation:** URDF/Xacro design, Gazebo integration, and environment setup.

- **Navigation and Control:** Motion control, joystick integration, and vision-based navigation behaviors.
- **Camera Integration:** Camera sensor setup and image topic handling.
- **Crop Stress Detection:** Image processing algorithms and ROS topic publication for crop health monitoring.

9. Conclusion

This project successfully demonstrates a simulated autonomous crop monitoring robot using ROS 2 and Gazebo. The system integrates robot modeling, simulation, camera-based perception, and navigation-related behaviors in a modular and extensible architecture. Experimental results confirm that the perception subsystem operates reliably and publishes real-time crop stress indicators, while the robot functions correctly within a simulated farm environment. The project aligns with the original proposal and highlights the potential of robotic and intelligent systems to support smallholder agriculture through early crop health monitoring.